

Resistors



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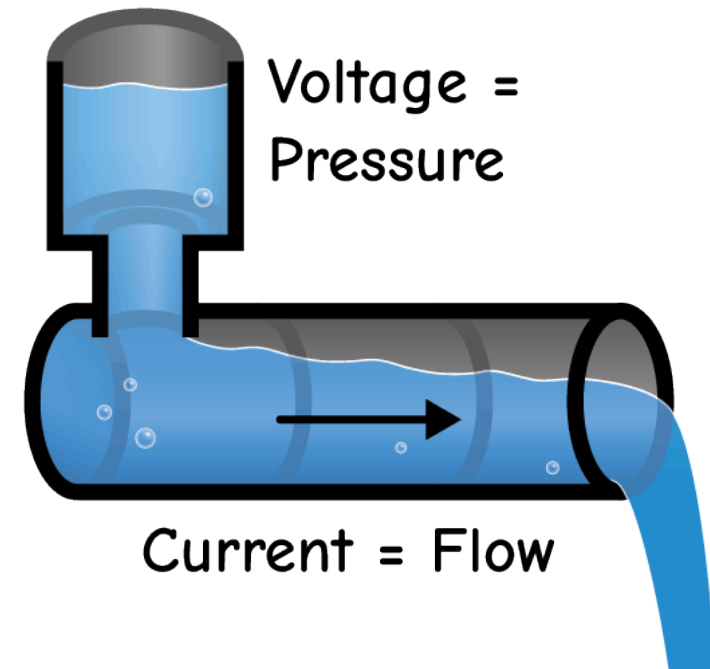
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Resistance

- Resistance is the opposition to the flow of electric current.
- Resistance has the symbol R and is measured in ohms (Ω).

Analogy

- Resistance is like a restriction in a water pipe.
- A greater restriction makes it harder for water to flow.
- Similarly, greater electrical resistance makes it harder for current to flow.



Equation for resistance of a wire

- Below is the equation for the resistance of a wire, some values we looked at in our practical others we did not

$$R = \rho \frac{L}{A}$$

Where:

R = Resistance (Ω)

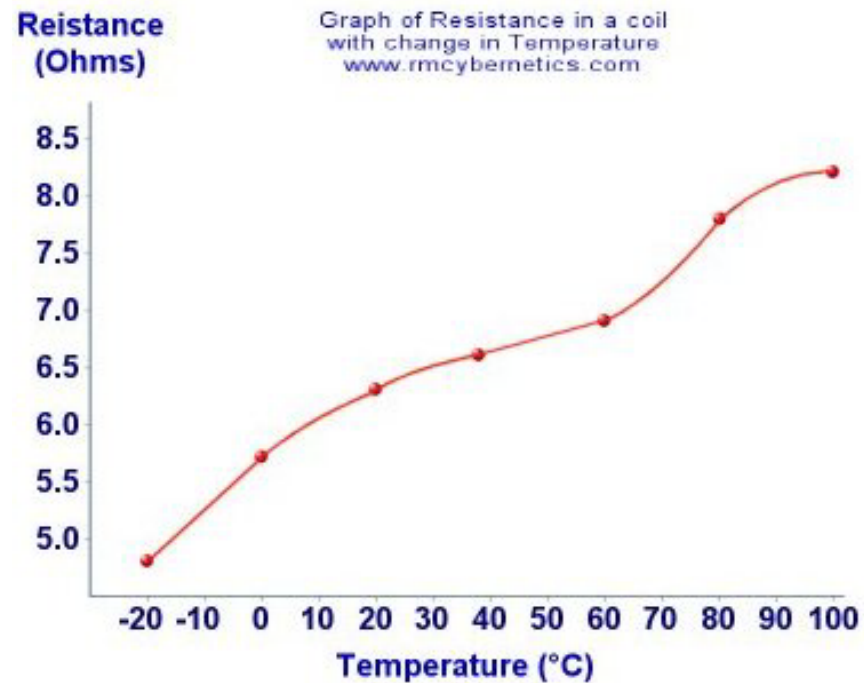
ρ = Resistivity of the material ($\Omega * m$)

L = Length of wire

A = Cross sectional area of the wire

Resistivity

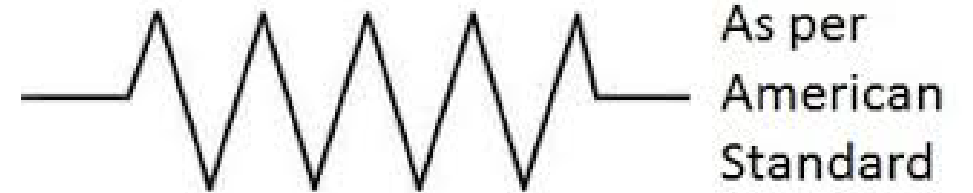
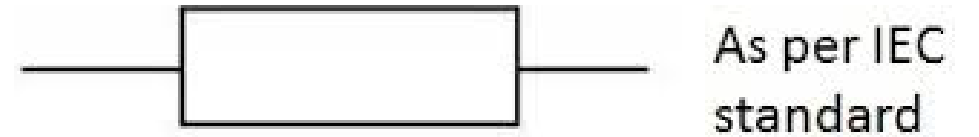
- The resistivity of the wire is predetermined by the material and the temperature of said material



Resistivity (Ω/m) of some materials		Less resistant
Carbon (graphene)	1×10^{-8}	
Silver	1.59×10^{-8}	
Copper	1.68×10^{-8}	
Gold	2.44×10^{-8}	
Calcium	3.36×10^{-8}	
Zinc	5.90×10^{-8}	
Lithium	9.28×10^{-8}	
Iron	1.0×10^{-7}	
Platinum	1.06×10^{-7}	
Tin	1.09×10^{-7}	More resistant
Lead	2.2×10^{-7}	
Carbon (graphite)	2.5×10^{-6} to 5.0×10^{-6}	
Sea water	2×10^{-1}	
Drinking water	2×10^1 to 2×10^3	
Silicon	6.40×10^2	
Deionized water	1.8×10^5	
Glass	10×10^{10} to 10×10^{14}	
Carbon (diamond)	1×10^{12}	
Hard rubber	1×10^{13}	
Sulfur	1×10^{15}	
Air	1.3×10^{16} to 3.3×10^{16}	
Teflon	10×10^{22} to 10×10^{24}	

What is a resistor

- If we take high resistance wire and wrap it round a core we get a resistor.
- They consist of a high resistance material wrapped around an insulating core



Function of a Resistor

Current Limiting

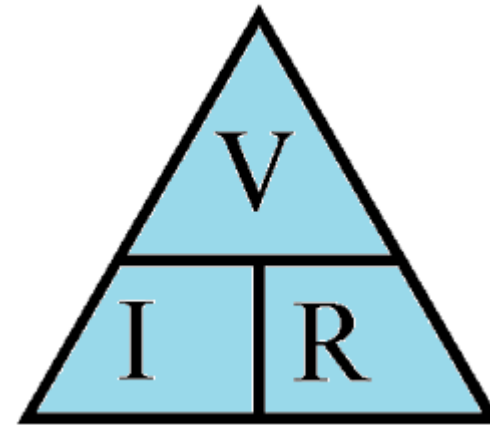
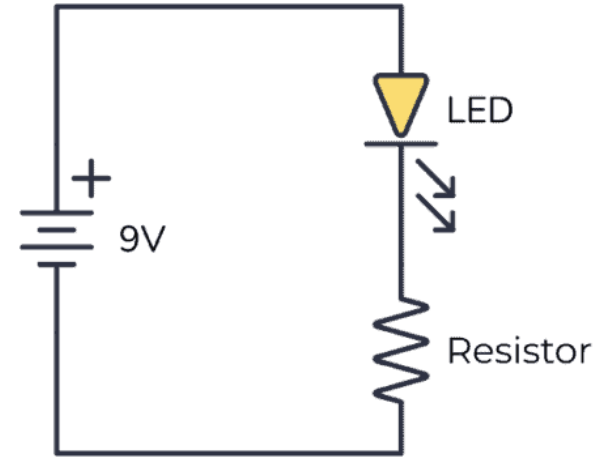
Voltage Division

Signal Conditioning

Heat Dissipation

Current Limiting

- **Purpose:** Protect components from excessive current.
- **Example:** LEDs need current limiting to prevent burning out.
- **How it Works:** The resistor restricts current according to Ohm's Law
- **Typical Application:** Resistor in series with an LED.
- **Key Point:** Without a resistor, sensitive components can be damaged.



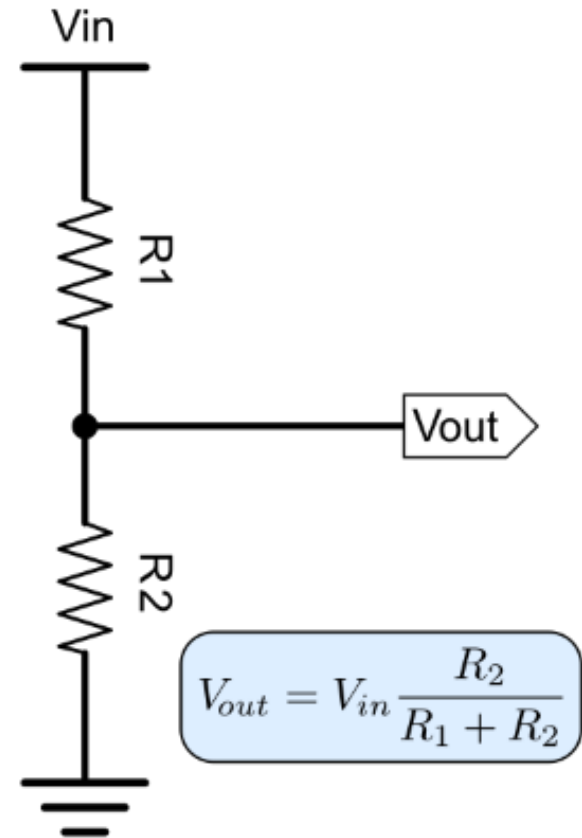
Heat Dissipation

- **Why it matters:** Resistors convert unwanted electrical energy into heat.
- **Power Rating:**
 - Small resistors: ¼ W – 1 W
 - Wirewound resistors: up to 100 W+
- **Applications:**
 - Used as heaters (to burn off excess power)
 - Protection in high-current circuits
- **Key Point:** Always choose a resistor with a power rating higher than expected dissipation.

$$P = VI = I^2R = \frac{V^2}{R}$$

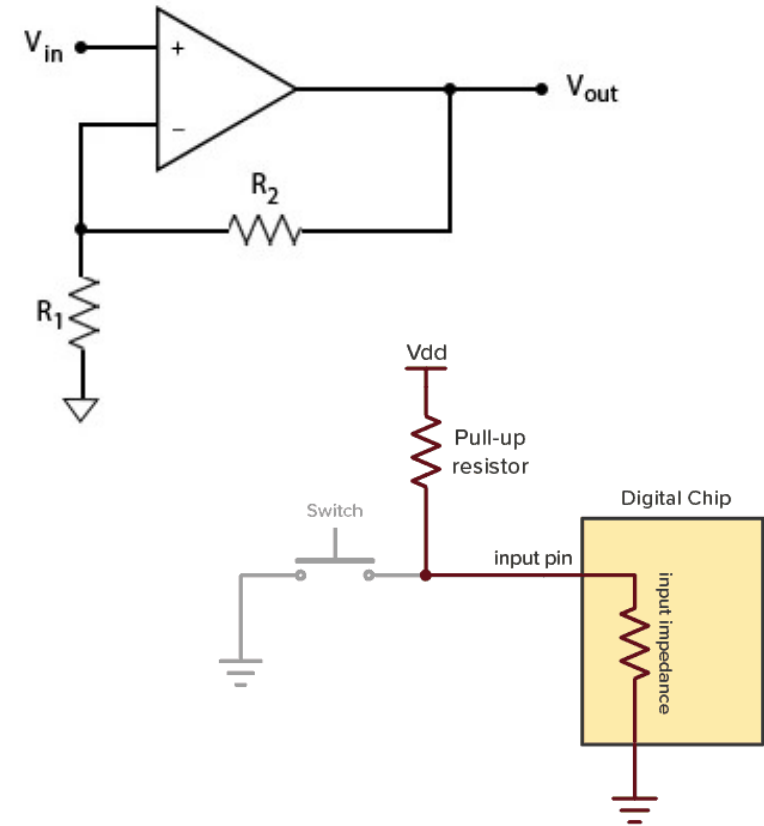
Voltage (potential) Divider

- **Purpose:** To split an input voltage into a smaller output voltage.
- **Circuit:** Two resistors in series across a supply.
- **Applications:**
 - Sensor circuits (e.g. LDRs, thermistors)
 - Reference voltages
 - Scaling signals for measurement



Signal Processing

- **Biasing:** Set operating points in amplifiers (e.g. transistor base bias).
- **Pull-up/Pull-down:** Ensure logic inputs default to a known state (0 or 1).
- **Filtering (with capacitors):** Form RC low-pass or high-pass filters. Control which frequencies are allowed through.
- **Impedance Control:** Match circuit stages to prevent signal loss or distortion.



Types of Resistors

- We can divide resistors 3 main ways:

- **Fixed**

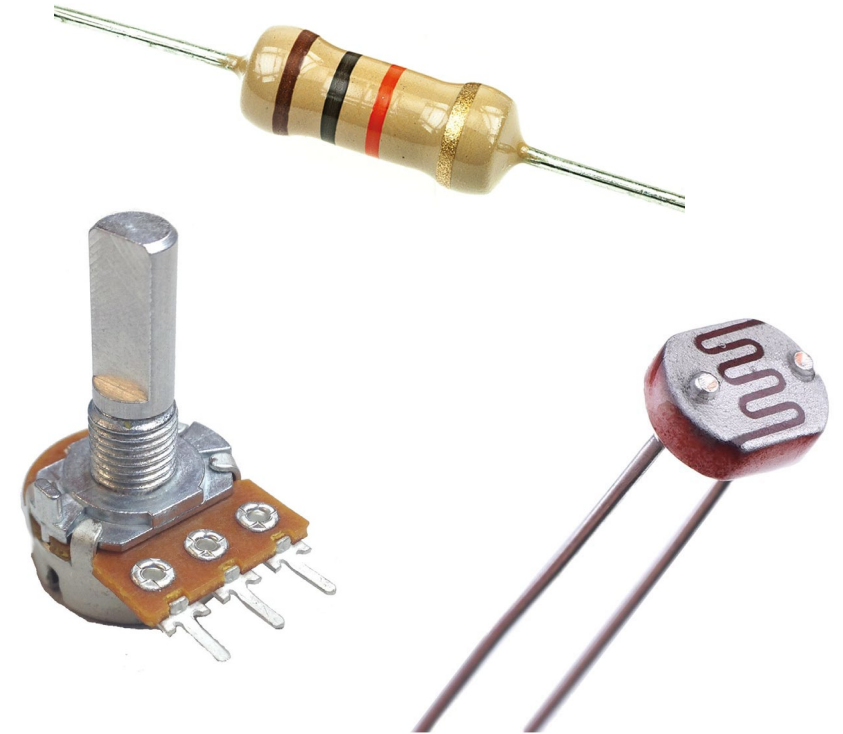
- Carbon Film
- Metal Film
- Wire Wound

- **Variable**

- Potentiometer
- Rheostats

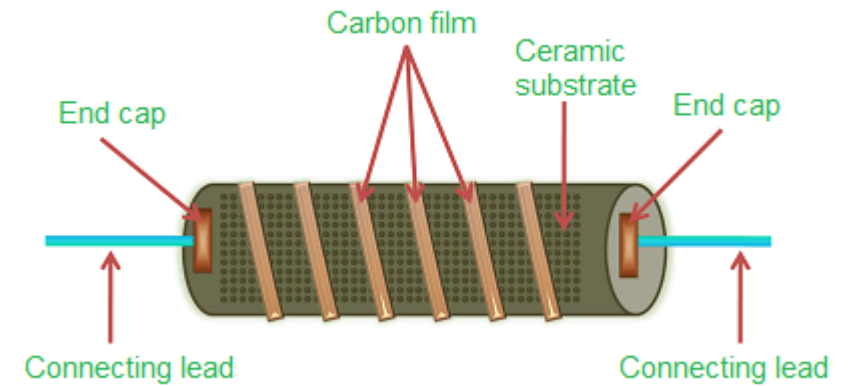
- **Special variable resistors**

- LDR
- Thermistor
- Varistor



Carbon Film Resistors

- **Construction:** A thin film of carbon deposited on an insulating substrate.
- **Resistance Value:** Controlled by the thickness and length of the carbon film.
- **Power Rating:** Typically, low to medium ($\frac{1}{4}$ W to 2 W).
- **Tolerance:** Around $\pm 5\%$ (standard), but can be tighter.



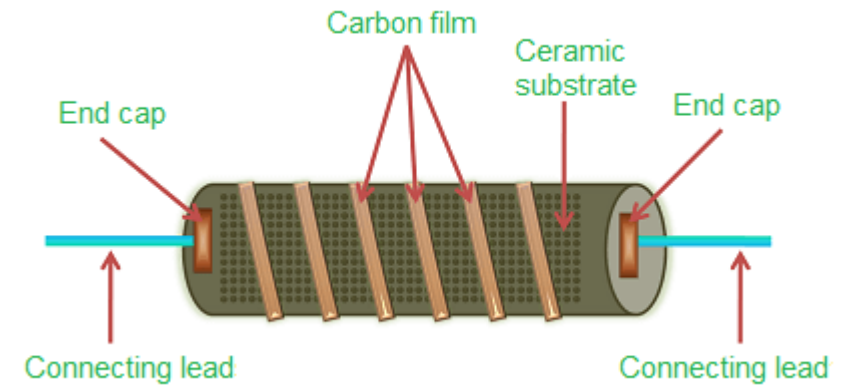
Carbon Film Resistors

- **Advantages:**

- Cheap and widely available
- Good stability for general use

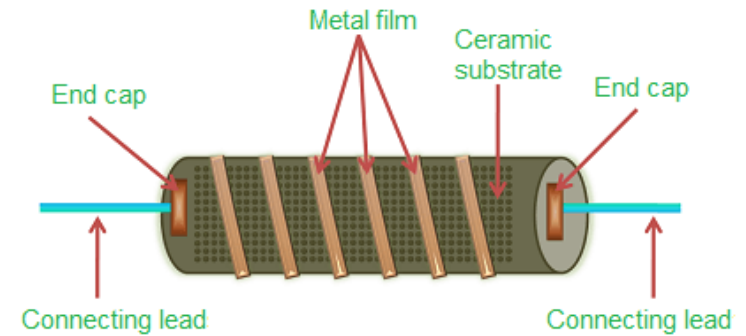
- **Limitations:**

- Higher noise than metal film resistors
- Not suitable for high precision applications



Metal Film Resistors

- **Construction:** Thin layer of metal (often nickel-chromium) deposited on a ceramic rod.
- **Resistance Value:** Adjusted by cutting a helical groove in the film.
- **Power Rating:** Typically low ($\frac{1}{8}$ W to 1 W).
- **Tolerance:** Very precise, often $\pm 1\%$ or better.



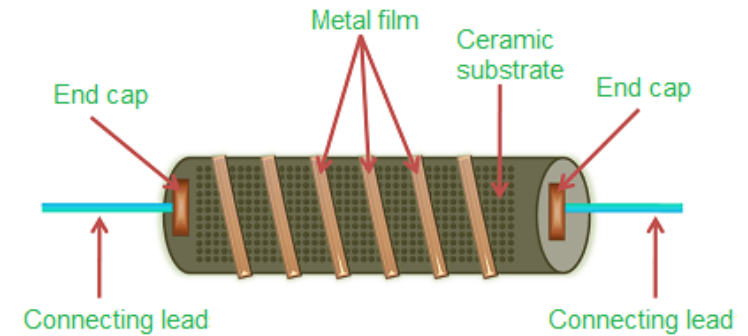
Metal Film Resistors

- **Advantages:**

- High accuracy and stability
- Low noise compared to carbon film
- Good temperature performance

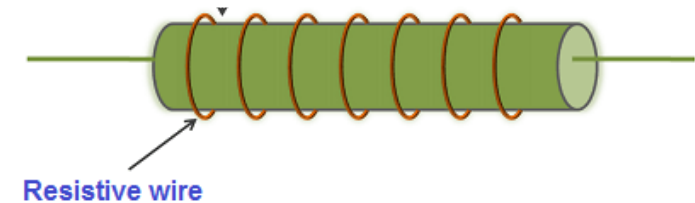
- **Limitations:**

- Slightly more expensive than carbon film
- Limited to low–medium power applications



Wire wound Resistors

- **Construction:** Resistive wire (usually nichrome) wound around a ceramic or fiberglass core.
- **Resistance Value:** Determined by the length and thickness of the wire.
- **Power Rating:** High – can handle several watts to hundreds of watts.
- **Tolerance:** Precise, typically $\pm 1\%$ or better.



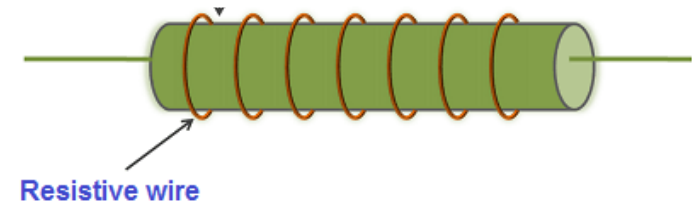
Wire wound Resistors

- **Advantages:**

- Excellent stability and accuracy
- Can dissipate large amounts of power
- Low temperature coefficient

- **Limitations:**

- Larger physical size
- Inductive properties (not ideal for high-frequency circuits)
- More expensive than film resistors



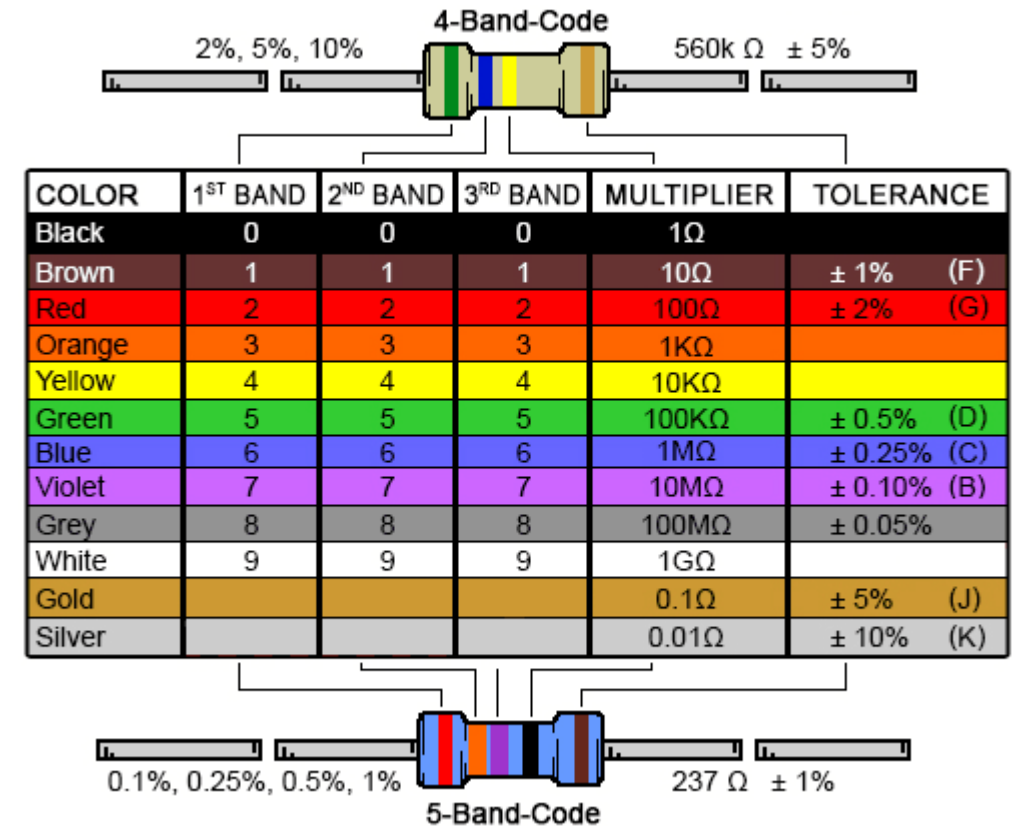
Identifying Resistors

- **Resistor Colour Code:**

- Each band represents a number or multiplier.
- Final band = tolerance (gold = $\pm 5\%$, silver = $\pm 10\%$).

- **Other Identifiers:**

- Marked values (printed on some precision resistors).
- Case size indicates power rating ($\frac{1}{4}$ W, $\frac{1}{2}$ W, etc.).



Identifying Resistors - Example

2%, 5%, 10%

4-Band-Code

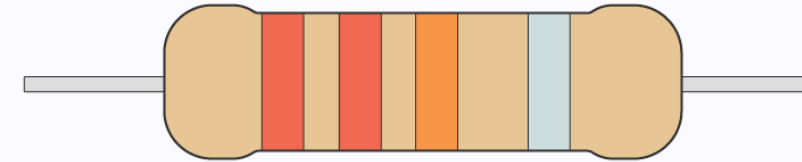
560k Ω $\pm$ 5%

COLOR	1 ST BAND	2 ND BAND	3 RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1 Ω	
Brown	1	1	1	10 Ω	$\pm$ 1% (F)
Red	2	2	2	100 Ω	$\pm$ 2% (G)
Orange	3	3	3	1K Ω	
Yellow	4	4	4	10K Ω	
Green	5	5	5	100K Ω	$\pm$ 0.5% (D)
Blue	6	6	6	1M Ω	$\pm$ 0.25% (C)
Violet	7	7	7	10M Ω	$\pm$ 0.10% (B)
Grey	8	8	8	100M Ω	$\pm$ 0.05%
White	9	9	9	1G Ω	
Gold				0.1 Ω	$\pm$ 5% (J)
Silver				0.01 Ω	$\pm$ 10% (K)

0.1%, 0.25%, 0.5%, 1%

5-Band-Code

237 Ω $\pm$ 1%



Red	Red	Orange	Silver
2	2	10 ³	±10%

22000 Ω $\pm$ 10%